

Telecom Customer Segmentation & Package Recommendation System

Business Report — Methodology, Results & Value

Dataset: Orange / BigML Telecom Churn — 3,333 subscribers • Overall churn rate 14.5%

1. Executive Summary

This project turns a raw telecom subscriber table into an actionable customer-intelligence system. We profiled 3,333 customers, engineered value and usage features, grouped customers into five business segments, built a recommendation engine that suggests a better package for each subscriber, and trained a churn-prediction model that flags at-risk customers before they leave — with an explainability layer that shows *why* each prediction is made.

Headline results

- **Churn prediction:** XGBoost achieves **0.924 ROC-AUC** and **87% recall** on churners (holdout test) — most leavers are caught.
- **Segmentation:** five clear segments; the **High Risk** segment (728 customers) churns at **32%** while paying the **highest bills** (\$72.80 avg) — the priority retention target.
- **Recommendations:** a cross-sell / upsell opportunity is flagged for **~40%** of subscribers.
- **Top churn drivers** (SHAP): total charges, voicemail plan, international calls, and customer-service calls.

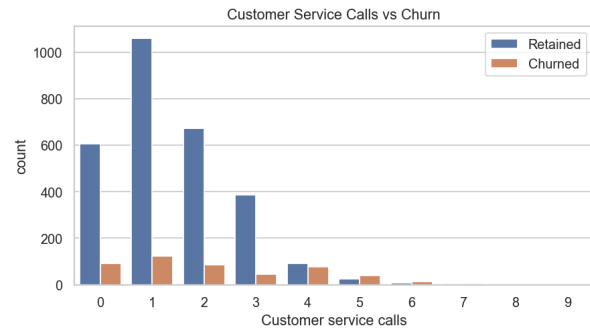
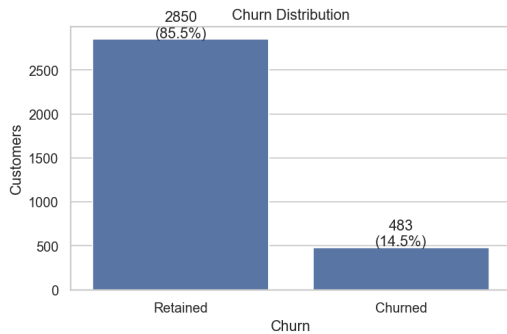
2. What Was Done

The work is organised as a reproducible 7-phase pipeline:

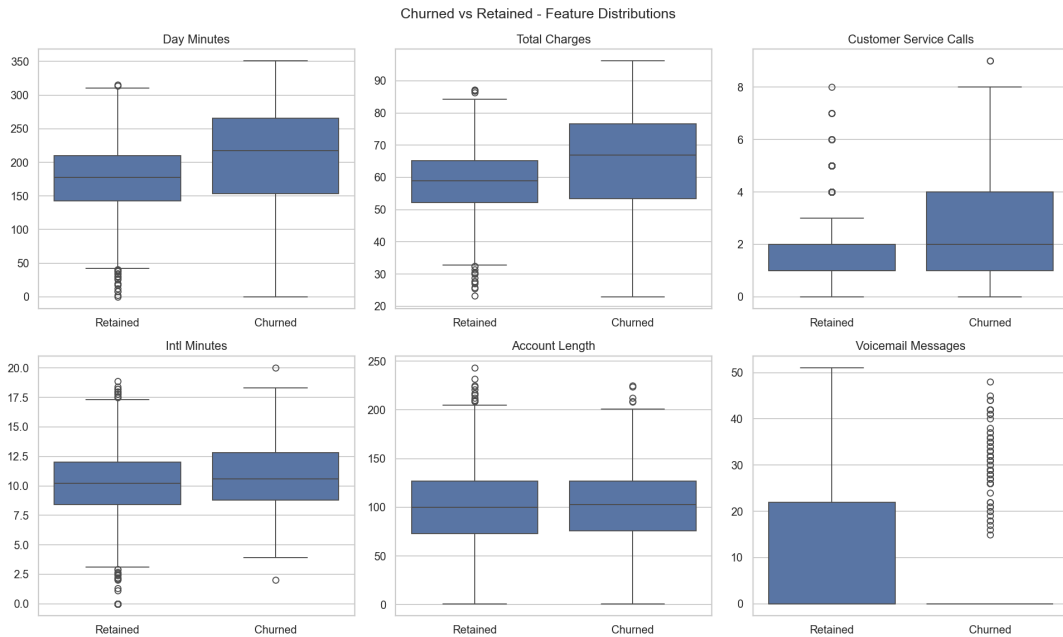
Phase	Activity	Output
1. EDA	Distributions, correlations, churn boxplots	7 charts + insights
2. Feature Engineering	Total usage/charges, intl ratio, value score, segments	Enriched dataset
3. Segmentation	K-Means + DBSCAN, profiled & named clusters	5 business segments
4. Recommendation	KNN — 5 most-similar customers per subscriber	Package per customer
5. Churn Prediction	KNN / RF / XGBoost + SMOTE + tuned CV	Best model (XGBoost)
6. Explainability	SHAP on the tuned model	Ranked churn drivers
7. Business Value	Translate analytics into operator actions	This report

3. Key Findings (EDA)

- **Heavy users** (top 10% by minutes) churn **47%** vs 14.5% overall.
- **Top payers** (top 10% by charges) churn **63%** — the most valuable customers are the most likely to leave.
- Customers with **≥4 service calls** churn **52%** — a clear distress signal.
- **International-plan** holders churn **42%** vs 11.5% without.



Churn distribution | Service calls vs churn



Churned vs retained customers across key features

4. Customer Segments

K-Means (with a DBSCAN cross-check) groups customers into five segments, each mapped to a business profile:

Segment	Customers	Churn	Avg Bill	Value Score
High Risk Users	728	32%	\$72.80	57.2
Premium Users	836	9%	\$60.10	46.6
Voice Heavy Users	509	10%	\$60.27	49.7
International Users	568	13%	\$44.97	34.7
Low Revenue Users	692	7%	\$55.91	43.1

High Risk Users (highlighted) are the standout concern: they pay the most yet churn at 32% — protecting them protects the most revenue.



5. Package Recommendation Engine

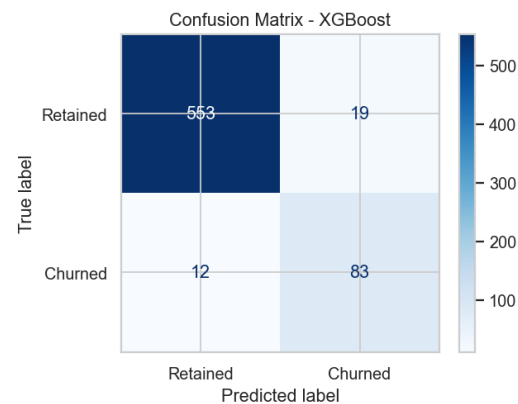
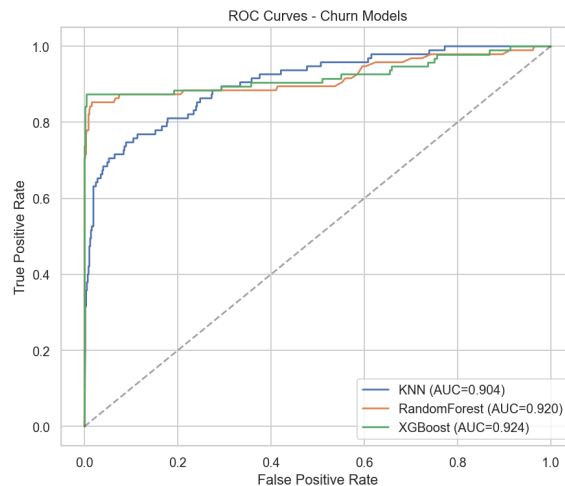
For every subscriber the engine finds the 5 most similar customers — by usage, charges, international calls, tenure and service calls — and recommends the package most common among them: “*Customers similar to you are using Package X.*” This surfaces a cross-sell / upsell opportunity for roughly **40% of the base**, and gives front-line staff a concrete next-best offer for each customer.

6. Churn Prediction Results

Three models were tuned with SMOTE (to handle the 14.5% imbalance) and Stratified 5-fold cross-validation, then scored on the untouched 20% holdout split.

Model	CV ROC-AUC	Test ROC-AUC	Test F1
KNN	0.880	0.904	0.560
RandomForest	0.915	0.920	0.876
XGBoost	0.925	0.923	0.843

XGBoost is the best model: **0.924 ROC-AUC**, 95% accuracy, and **87% recall on churners** — it catches the large majority of customers who are about to leave.



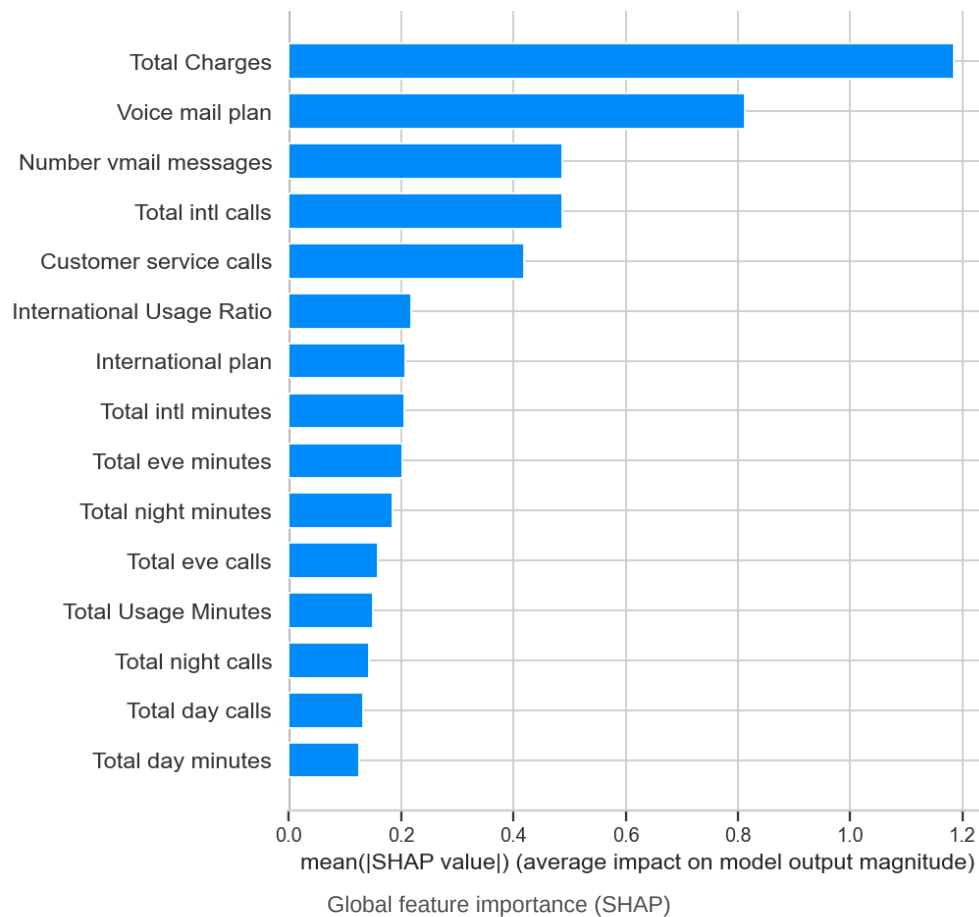
ROC curves | Confusion matrix (best model)

7. Why Customers Churn (SHAP)

SHAP explains the model's predictions. The strongest churn drivers are:

Rank	Driver	Impact (mean SHAP)
1	Total Charges	1.184
2	Voice mail plan	0.812
3	Number vmail messages	0.488
4	Total intl calls	0.486
5	Customer service calls	0.418
6	International Usage Ratio	0.218

SHAP Global Feature Importance



8. Business Value

The pipeline converts raw subscriber data into five concrete operator capabilities:

- **Identify high-value customers** — the Customer Value Score and segments rank the base by worth.
- **Recommend better packages** — the KNN engine gives every subscriber a next-best-offer, with upsell flags.
- **Reduce churn** — every subscriber gets a churn probability and risk band, so retention teams act *before* customers leave.
- **Increase ARPU** — move Low-Revenue users onto fitting paid packages and upsell Premium / International users.
- **Sharper retention campaigns** — target the **High-Risk & High-Value** overlap, where saved revenue per customer is greatest.

Recommended actions

- Launch a proactive save-desk for High-Risk customers with ≥ 3 service calls.
- Audit the international plan — its holders churn $\sim 4\times$ the base; revisit pricing / quality.
- Feed the scored dataset (`telecom_scored.csv`) into a Power BI dashboard for ongoing monitoring.

All figures in this report are produced by the reproducible pipeline; the scored dataset is Power BI-ready.